National University of Computer & Emerging Sciences - NUCES - Karachi

FAST School of Computing

AI LAB 11 Tasks

TASK # 1

In the example code in lab manual, customer segmentation was performed using only two features. Now, use all available features except customer_id to implement K-Means clustering. Perform the clustering twice — once without feature scaling, and once with scaling applied to all features except age. Compare the results and comment on the differences and insights observed from these changes.

TASK # 2

A transportation company wants to optimize its routes and fleet by categorizing different types of vehicles based on their usage patterns. The company has data on several features such as: vehicle_serial_no, mileage, fuel_efficiency, maintenance_cost, and vehicle_type. The goal is to create segments of vehicles based on these attributes to help with fleet management. Implement K-Means clustering to group vehicles using all the features. Perform the clustering twice: once without scaling the features and once with scaling applied to all features (except for vehicle_type, which is categorical). Analyze and compare the results, focusing on how the scaling affects the clustering output.

Sample Data:

```
# Example vehicle data
data = {
   'vehicle_serial_no': [5, 3, 8, 2, 4, 7, 6, 10, 1, 9],
   'mileage': [150000, 120000, 250000, 80000, 100000, 220000, 180000, 300000, 75000,
280000],
   'fuel_efficiency': [15, 18, 10, 22, 20, 12, 16, 8, 24, 9],
   'maintenance_cost': [5000, 4000, 7000, 2000, 3000, 6500, 5500, 8000, 1500, 7500],
   'vehicle_type': ['SUV', 'Sedan', 'Truck', 'Hatchback', 'Sedan', 'Truck', 'SUV', 'Truck',
'Hatchback', 'SUV']
}
# Create a DataFrame
df = pd.DataFrame(data)
```

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TASK # 3

The academic affairs department at FAST NUCES Karachi is looking to identify distinct groups of students based on their academic engagement and performance. The department has access to anonymized student data containing the attributes student_id, GPA, study_hours (average weekly study hours), and attendance_rate (percentage of classes attended).

The goal is to group students into meaningful clusters that can help tailor academic support programs, such as extra tutoring, mentoring sessions, or motivation workshops.

You are required to perform unsupervised learning using K-Means clustering on the student dataset. Do following :

- Feature Selection and Scaling: Use the following features for clustering: GPA, study_hours, and attendance_rate.
- Apply appropriate feature scaling before clustering.
- Determine Optimal Number of Clusters (K): Use the ELbow method to determine the optimal number of clusters (K) in the range of 2 to 6.
- Perform Clustering: Apply K-Means using the optimal K and assign a cluster label to each student.
- Visualization: Create a scatter plot to visualize the clusters using study_hours and GPA as the axes.
- Color each point based on its cluster.
- Add an informative title and labels for clarity.
- Deliverables: Display the final dataset showing student IDs along with their assigned cluster.
- Present the scatter plot that illustrates the clustering result.